AMCS

Practice 3 Recurrent Problems 2. The Josephus Problem. Generalization of the Josephus problem. The repertoire method.

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The Josephus Problem

$J(n) =$ the original position of the last surviving person.

For the standard Josephus problem, we can use:

Recurrence:	$J(1) = 1, \quad J(2n) = 2J(n) - 1, \quad J(2n+1) = 2J(n) + 1,$
Closed form:	$n = 2^m + l, \quad J(n) = 2l + 1,$
Binary form:	$(1b_{m-1} \ldots b_0)_2 \to (b_{m-1} \ldots b_0 1)_2.$

Example: Find $J(6)$

Example: Find $J(10)$

1. Compute $J(7)$ for the standard Josephus problem, where every second person is eliminated. Use the closed form.
2. Compute $J(18)$ for the standard Josephus problem. Then verify the answer using the binary form.
3. Compute $J(45)$, $J(73)$, and $J(100)$ using the closed form

$$J(2^m + l) = 2l + 1.$$

4. A group of 26 students stands in a circle. Starting from student 1, every second remaining student is eliminated until only one student remains. Which student survives?
5. A game has 58 players arranged in a circle and numbered from 1 to 58. Every second player is removed from the game. Find the number of the last remaining player using the Josephus closed form and check your answer using binary representation.

6. Find $f(47)$.

$$\begin{cases} f(1) = 2, \\ f(2) = 5, \\ f(2n) = 3f(n) + 1, \\ f(2n+1) = 3f(n) + 4. \end{cases}$$

7. Find $f(121)$.

$$\begin{cases} f(1) = 1, \\ f(2) = 7, \\ f(3n) = 10f(n) + 5, \\ f(3n+1) = 10f(n) + 1, \\ f(3n+2) = 10f(n) + 8. \end{cases}$$

8. Find $f(341)$.

$$\begin{cases} f(1) = 3, \\ f(2) = 12, \\ f(3) = 7, \\ f(4n) = 7f(n) - 2, \\ f(4n+1) = 7f(n) - 3, \\ f(4n+2) = 7f(n) + 4, \\ f(4n+3) = 7f(n) + 8. \end{cases}$$

9. In a coding contest, the score $S(n)$ is defined recursively by

$$\begin{cases} S(1) = 4, \\ S(2n) = 2S(n) + 3, \\ S(2n+1) = 2S(n) + 7. \end{cases}$$

Find $S(93)$.

10. A computer process reduces an integer recursively according to the following rule:

$$\begin{cases} F(1) = 5, \\ F(2) = 9, \\ F(3) = 4, \\ F(3n) = 4F(n) + 2, \\ F(3n+1) = 4F(n) + 5, \\ F(3n+2) = 4F(n) + 7. \end{cases}$$

Find $F(245)$.

Generalized Josephus Problem

The original Josephus recurrence is

$$J(1) = 1,$$

$$J(2n) = 2J(n) - 1,$$

$$J(2n+1) = 2J(n) + 1.$$

The **Generalized Josephus Problem** replaces the constants $1, -1, 1$ by parameters α, β, γ:

$$\boxed{\begin{aligned} f(1) &= \alpha, \\ f(2n) &= 2f(n) + \beta, \\ f(2n+1) &= 2f(n) + \gamma. \end{aligned}}$$

The original Josephus function is a special case with

$$\boxed{\alpha = 1, \qquad \beta = -1, \qquad \gamma = 1.}$$

Example 1

Given

$$f(1) = 2,$$

$$f(2n) = 2f(n) + 3,$$

$$f(2n+1) = 2f(n) + 5,$$

find $f(4)$.

Instead of repeatedly using the recursive formula, we can use a **closed formula.**

First write

$$\boxed{n = 2^k + l},$$

where 2^k is the largest power of 2 not greater than n, and

$$\boxed{0 \leq l < 2^k}.$$

Then

$$\boxed{f(n) = 2^k\alpha + (2^k - 1 - l)\beta + l\gamma.}$$

Example 2 — Closed Formula

Given

$$f(1) = 3,$$

$$f(2n) = 2f(n) - 1,$$

$$f(2n+1) = 2f(n) + 2,$$

find $f(10)$.

Problem 1

Given

$$f(1) = 4,$$

$$f(2n) = 2f(n) + 2,$$

$$f(2n+1) = 2f(n) + 5,$$

find

$$\boxed{f(7)}$$

using the recursive formula.

Problem 2

Given

$$f(1) = 2,$$

$$f(2n) = 2f(n) - 3,$$

$$f(2n+1) = 2f(n) + 4,$$

find

$$\boxed{f(12)}$$

using the closed formula.

3.

Given

$$f(1) = 2,$$

$$f(2n) = 2f(n) + 1,$$

$$f(2n+1) = 2f(n) + 3.$$

Find

$$f(4).$$

5.

Given

$$f(1) = 4,$$

$$f(2n) = 2f(n) + 2,$$

$$f(2n+1) = 2f(n) + 5.$$

Find

$$f(10)$$

using the closed formula

$$f(n) = 2^k\alpha + (2^k - 1 - l)\beta + l\gamma,$$

where

$$n = 2^k + l.$$

6. Space Mission

A spacecraft collects data during a sequence of missions. The amount of collected data is modeled by

$$f(1) = 10.$$

For an even-numbered mission,

$$f(2n) = 2f(n) - 4,$$

and for an odd-numbered mission,

$$f(2n+1) = 2f(n) + 8.$$

Find the amount of data collected at mission

$$n = 25$$

using the closed formula.

7. Computer Game Challenge

A character starts with **8 experience points**. The experience function satisfies

$$f(1) = 8.$$

For an even-numbered stage, the new value is twice the corresponding previous value minus 5:

$$f(2n) = 2f(n) - 5.$$

For an odd-numbered stage, the new value is twice the corresponding previous value plus 10:

$$f(2n+1) = 2f(n) + 10.$$

Find the character's experience points at stage

$$n = 42.$$

First identify α, β, γ, then find k and l, and use the closed formula.

8. AI Training Model

An AI model is trained recursively on datasets of different sizes. For the smallest dataset, its performance score is

$$f(1) = 12.$$

When the dataset index is even, its score is twice the score of the corresponding smaller dataset, with a penalty of 7 points. When the index is odd, its score is twice the score of the corresponding smaller dataset, with a bonus of 15 points.

Find the performance score

$$f(100).$$

Write the recurrence relation first. Then identify

$$\alpha, \qquad \beta, \qquad \gamma,$$

represent 100 as

$$100 = 2^k + l,$$

and use the closed formula

$$\boxed{f(n) = 2^k\alpha + (2^k - 1 - l)\beta + l\gamma.}$$

Repertoire Method

For the **Repertoire Method:**

$$\boxed{f(n) = A(n)\alpha + B(n)\beta + C(n)\gamma}$$

with

$$\boxed{A(n) = 2^k}$$

$$\boxed{A(n) - B(n) - C(n) = 1}$$

$$\boxed{A(n) + C(n) = n}$$

and finally

$$\boxed{A(n) = 2^k, \qquad B(n) = 2^k - 1 - l, \qquad C(n) = l.}$$

Example 1

Use the repertoire function

$$f(n) = 1.$$

Find α, β, γ.

Example 2

Use the repertoire function

$$f(n) = n.$$

Find α, β, γ.

Example 3

Let

$$n = 11.$$

Find $A(11), B(11), C(11)$.

Problem 1

For the repertoire function

$$f(n) = 1,$$

use

$$f(1) = \alpha, \qquad f(2n) = 2f(n) + \beta, \qquad f(2n+1) = 2f(n) + \gamma$$

to find

$$\boxed{\alpha, \beta, \gamma}.$$

Problem 3

Let

$$n = 25.$$

Using

$$n = 2^k + l,$$

find

$$\boxed{A(25), \quad B(25), \quad C(25)}$$

where

$$A(n) = 2^k, \qquad B(n) = 2^k - 1 - l, \qquad C(n) = l.$$